KPI Forecasting and Monitoring Diagnostics

Synthetic Insurance and Operational KPIs | R

This report builds a forecasting and monitoring framework for synthetic insurance and operational KPIs. The workflow combines ARIMA forecasting, AIC-selected regression models, rolling forecast evaluation, residual diagnostics and anomaly alerts to produce a compact business monitoring report.

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1 Executive summary

This project develops an end-to-end KPI forecasting and monitoring workflow in R. The dataset is synthetic, but it is designed to resemble a monthly insurance and operational performance setting. It includes claims count, earned premium, lapse rate, loss ratio and an operating margin proxy, together with supporting drivers such as policy exposure, inflation, seasonality and economic stress.

The modelling framework has two parts. First, ARIMA models are fitted to each KPI and used for twelve-month forecasts with prediction intervals. Second, AIC-selected regression models are used to link KPI movements to business drivers. The regression component is included not only for prediction, but also to show how operational and financial movements can be explained in business terms.

The models are evaluated using expanding-window one-step-ahead rolling forecasts. This avoids the overly optimistic interpretation that can result from fitting a model once on the full dataset. The forecast evaluation reports MAE, RMSE, MAPE, bias and 95% interval coverage. Coverage is close to the nominal target across the five KPIs, ranging from 94.4% to 97.2%.

The diagnostics section checks residual behaviour using summary statistics and Ljung-Box tests. The loss-ratio model is shown visually as a representative diagnostic example because loss ratio is a key insurance performance KPI. The same diagnostic process is applied across all forecasted KPIs.

The monitoring section converts the modelling outputs into a dashboard-style workflow. It reports current KPI movements, rolling threshold breaches and forecast interval breaches. The result is a compact forecasting and monitoring report that can be interpreted in either an insurance analytics context or a broader finance and business analytics context.

2 Data and KPI design

The project uses a synthetic monthly dataset from January 2019 to December 2025, giving 84 observations. The data are constructed to represent a realistic monitoring environment rather than a purely random sample. The series include trend, seasonality, exposure effects, economic stress effects and occasional shock behaviour.

The five main KPIs are:

- **Claims Count:** a volume measure representing the number of claims in each month.
- **Earned Premium / Revenue:** a financial volume measure used both as an insurance premium measure and a broader revenue proxy.
- **Lapse Rate:** a policyholder behaviour KPI showing the proportion of business lapsing in a period.
- **Loss Ratio:** a core insurance performance KPI measuring claims cost relative to earned premium.
- **Operating Margin Proxy:** a finance-oriented profitability measure that responds to claim

and expense pressure.

The dataset also includes modelling drivers such as policy count, inflation index, economic stress, month-of-year effects and lagged KPI values. This allows the project to support both forecasting and explanatory analytics.

Table 1: Dataset summary

Metric	Value
Data start	2019-01-01
Data end	2025-12-01
Monthly observations	84
Average policy count	12,581
Average earned premium / revenue	1,258,208
Average claims count	463
Average loss ratio	44.64%
Average lapse rate	4.12%
Average operating margin proxy	32.63%

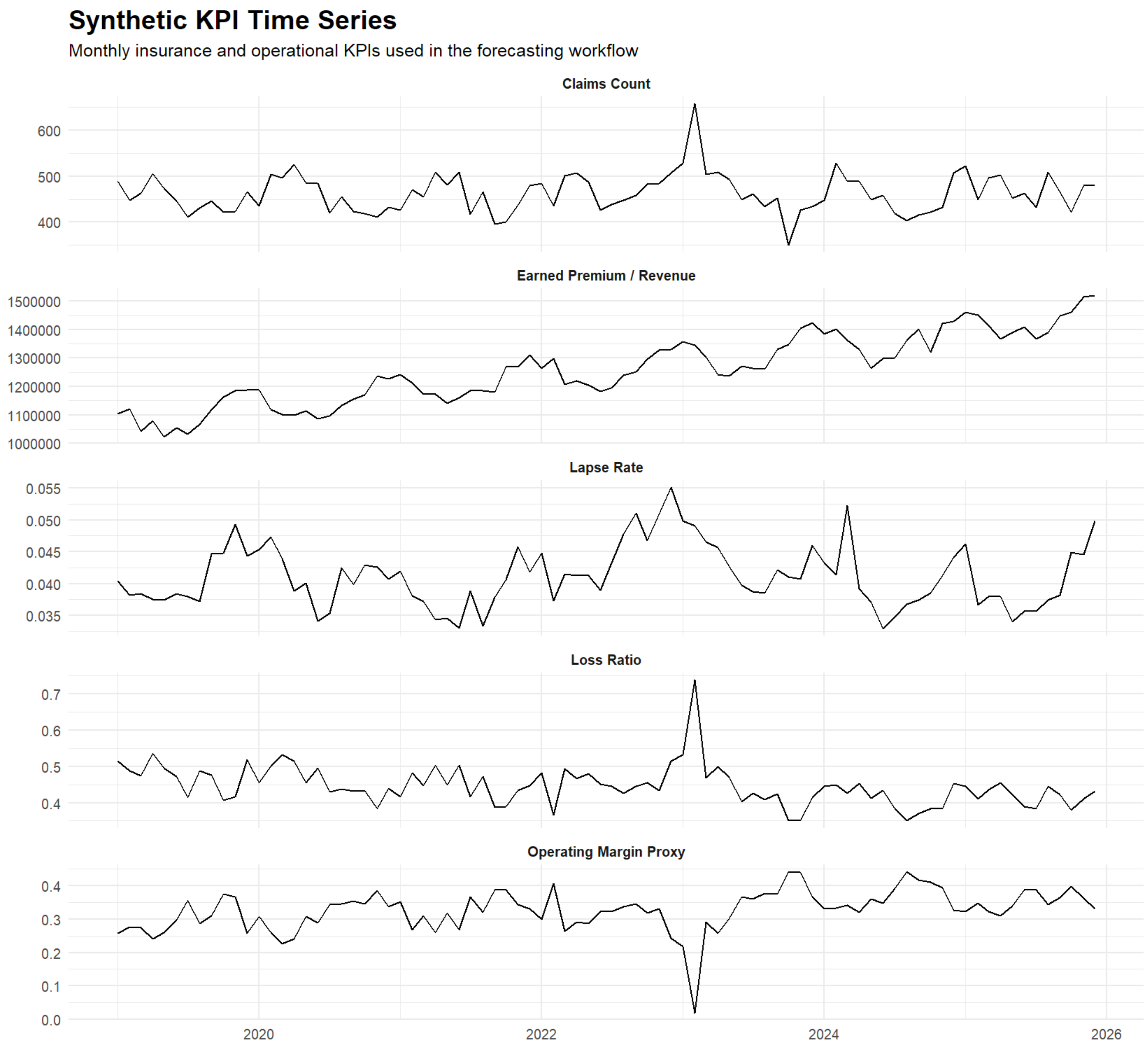

Figure 1: Synthetic monthly KPI series. The dataset combines insurance KPIs with broader operational and financial performance measures.

The time series plot shows the intended structure of the synthetic data. Earned premium has a clear upward trend, while claims count, loss ratio and lapse rate show more short-term volatility. The shock around 2023 is visible in claims count, loss ratio and operating margin, creating a realistic stress period for the monitoring system to detect.

3 Methodology

3.1 ARIMA forecasting

Each KPI is modelled using an ARIMA specification. ARIMA models are appropriate as a baseline because they can capture autoregressive structure, differencing behaviour, seasonality and moving-average effects in a univariate time series. For a KPI series y_t, a simplified ARIMA structure can be written as:

$$\phi(B)(1-B)^d y_t = c + \theta(B)\varepsilon_t, \tag{1}$$

where B is the backshift operator, d is the order of differencing, $\phi(B)$ captures autoregressive terms, $\theta(B)$ captures moving-average terms and ε_t is the residual innovation.

Model selection is based on the Akaike Information Criterion:

$$\text{AIC} = 2k - 2\log(\hat{L}), \tag{2}$$

where k is the number of estimated parameters and $\hat{L}$ is the maximised likelihood. AIC balances goodness of fit against model complexity, so it is useful for selecting parsimonious time-series specifications [1].

Table 2: Selected ARIMA models

KPI	Selected ARIMA model	AIC	BIC	Innovation variance
Claims Count	ARIMA(1,0,0)	853.88	861.17	1,447
Earned Premium / Revenue	ARIMA(0,0,0)(1,1,0)[12]	1670.75	1677.58	597,351,798
Lapse Rate	ARIMA(3,0,0)(0,1,2)[12]	-596.74	-583.08	0.00001
Loss Ratio	ARIMA(2,1,1)	-257.10	-247.42	0.00245
Operating Margin Proxy	ARIMA(2,1,1)	-251.65	-241.97	0.00262

The selected ARIMA models vary across KPIs, which is expected. Earned premium requires seasonal differencing, while loss ratio and operating margin are better represented by differenced ARIMA specifications. Claims count is represented by a simpler autoregressive model.

3.2 Regression driver models

Regression models are included to connect KPI movements to interpretable business drivers. For a KPI y_t, the general regression form is:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_p x_{p,t} + u_t, \tag{3}$$

where the x variables represent drivers such as month-of-year effects, policy count, economic stress, inflation and lagged KPI values.

AIC is again used to select parsimonious specifications. This avoids retaining unnecessary predictors simply because they are available. The purpose is not to build the largest possible regression, but to produce a driver model that is useful and interpretable.

The operating-margin model is handled carefully. Same-period loss ratio and same-period expense ratio are not used as inputs because those variables mechanically define operating margin. Including them would make the regression look artificially strong and would amount to reconstructing a formula rather than modelling a predictive relationship. Instead, the selected model uses lagged and external drivers.

Table 3: Selected regression models

KPI	Selected regression drivers	AIC	Adjusted R^2	Residual SE
Claims Count	Month effects, policy count and economic stress	818.68	0.476	30.70724
Earned Premium / Revenue	Month effects, policy count, economic stress and inflation index	1898.28	0.972	20,402
Lapse Rate	Month effects, economic stress and lagged lapse rate	-734.01	0.704	0.00266
Loss Ratio	Month effects, economic stress and inflation index	-284.93	0.474	0.03981
Operating Margin Proxy	Month effects, economic stress, inflation index, lagged loss ratio and lagged operating margin	-287.59	0.594	0.03881

The regression results are plausible for a synthetic business dataset. Earned premium is strongly explained by exposure and inflation drivers. Lapse rate is linked to economic stress and its own lag. Operating margin is explained with a moderate adjusted R^2, which is more credible than a mechanically perfect fit.

3.3 Rolling forecast evaluation

The forecasting models are evaluated using expanding-window one-step-ahead rolling forecasts. At each evaluation date, the model is fitted using the data available up to that point and then used to forecast the next month. This produces a sequence of out-of-sample forecast errors.

The main error metrics are:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} |y_t - \hat{y}_t|, \tag{4}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} (y_t - \hat{y}_t)^2}, \tag{5}$$

$$\text{MAPE} = \frac{100}{n}\sum_{t=1}^{n} \left|\frac{y_t - \hat{y}_t}{y_t}\right|. \tag{6}$$

Bias is calculated as the average forecast error. Prediction interval coverage is calculated as the proportion of actual observations falling inside the model's 95% prediction interval.

4 Forecasting results

The forecast output is summarised using a compact chart showing historical values, twelve-month forecasts and 95% prediction intervals for all five KPIs.

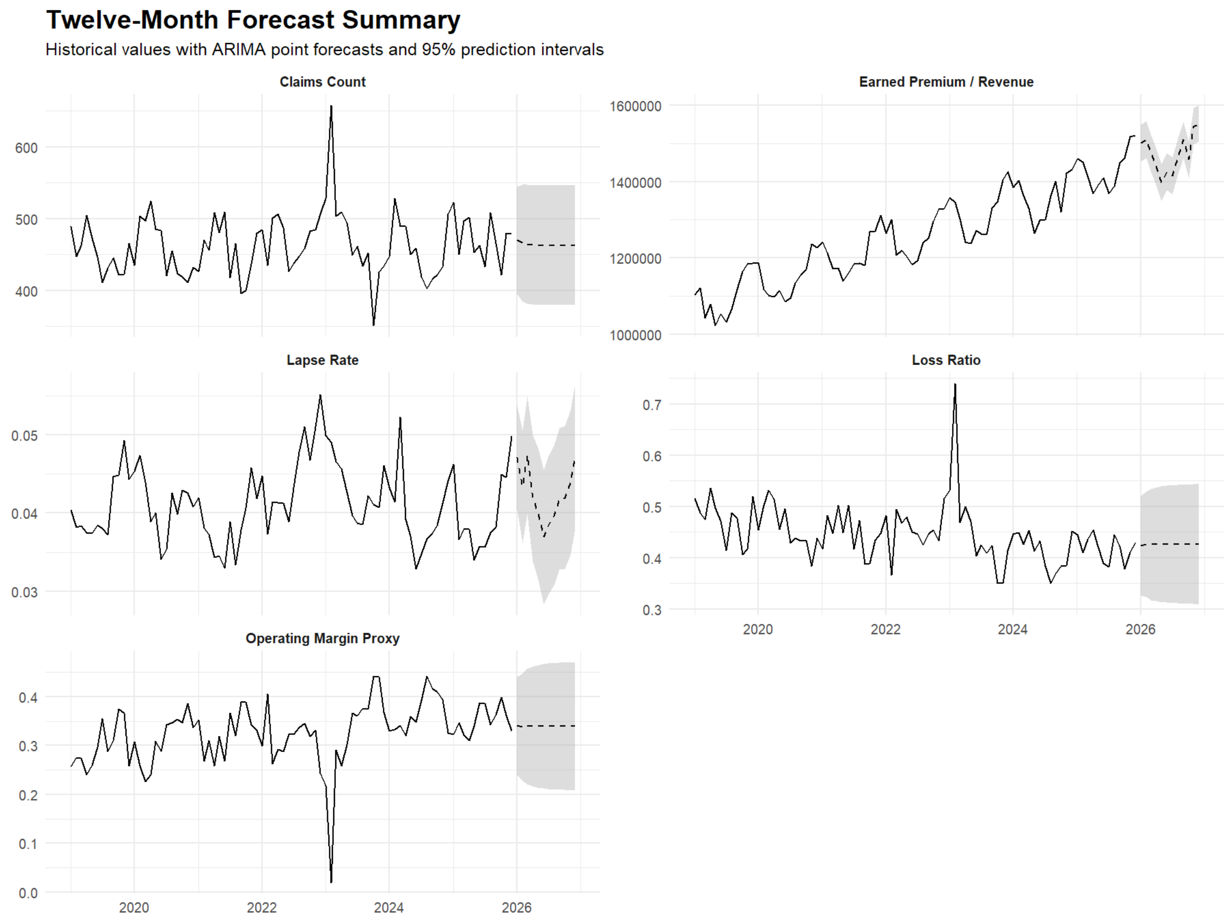

Figure 2: Twelve-month ARIMA forecast summary. Historical values are shown with point forecasts and 95% prediction intervals for each KPI.

The forecasts behave differently across the five KPIs. Earned premium continues its upward trend, while claims count remains relatively volatile. Lapse rate, loss ratio and operating margin have bounded rate-style dynamics, so the interpretation should focus on the range of plausible future values rather than only on the central point forecast.

Table 4: Rolling one-step-ahead forecast performance

KPI	Observations	MAE	RMSE	MAPE	Bias	95% interval coverage
Claims Count	36	32.4	45.2	6.8%	2.3	94.4%
Earned Premium / Revenue	36	19,718	26,326	1.5%	1,773	94.4%
Lapse Rate	36	0.26 p.p.	0.33 p.p.	6.3%	-0.04 p.p.	97.2%
Loss Ratio	36	3.89 p.p.	5.76 p.p.	8.8%	-0.93 p.p.	94.4%
Operating Margin Proxy	36	4.12 p.p.	5.89 p.p.	42.8%	0.99 p.p.	94.4%

For claims count and earned premium, MAE, RMSE and bias are reported in the natural units of the KPI. For lapse rate, loss ratio and operating margin, MAE, RMSE and bias are reported in percentage-point terms. MAPE remains a relative percentage error for all KPIs.

The interval coverage results are close to the nominal 95% target. This suggests that the forecast intervals are reasonably calibrated for the synthetic data. The operating margin MAPE is high, but MAPE can be unstable for margin and rate series, especially when the denominator is affected by shocks or low values. For operating margin, the MAE and RMSE in percentage-point terms are more interpretable.

5 Forecast error monitoring

Rolling forecast errors are monitored using a six-month rolling RMSE. This provides a practical way to identify periods where forecast quality deteriorates. A model can have acceptable average performance but still show periods of instability, so rolling diagnostics are useful for monitoring model risk over time.

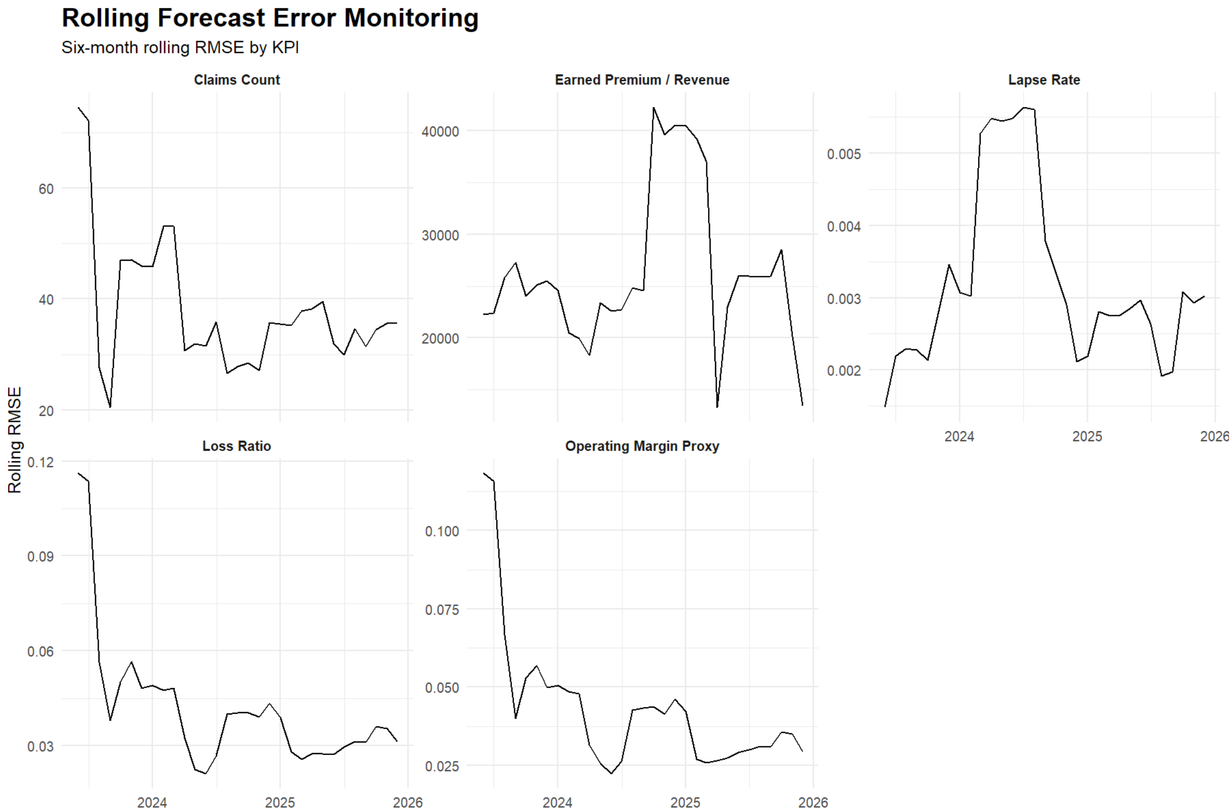

Figure 3: Six-month rolling RMSE by KPI. Rolling forecast errors help identify periods where model performance deteriorates or improves.

The rolling RMSE chart shows that forecast errors vary over time. Earned premium has a visible error spike around periods where the level of the series changes. Claims count and lapse rate also show periods of higher forecast error. This is useful from a monitoring perspective because forecast error instability can be treated as a review trigger rather than only as a model evaluation statistic.

6 Model health diagnostics

Residual diagnostics were used to check whether the fitted ARIMA models left behind material autocorrelation. The Ljung-Box test is used as a portmanteau test for residual autocorrelation [5]. A low p-value indicates that the residuals may still contain time-series structure not captured by the model.

Table 5: Model health summary

KPI	Selected model	Ljung-Box p-value	Residual mean	Residual SD	Health status
Claims Count	ARIMA(1,0,0)	0.0014	-0.2	37.8	Review residual autocorrelation
Earned Premium / Revenue	ARIMA(0,0,0)(1,1,0)[12]	0.4692	-381	22,442	No material residual autocorrelation
Lapse Rate	ARIMA(3,0,0)(0,1,2)[12]	0.3678	0.01 p.p.	0.30 p.p.	No material residual autocorrelation
Loss Ratio	ARIMA(2,1,1)	0.2253	-0.76 p.p.	4.80 p.p.	No material residual autocorrelation
Operating Margin Proxy	ARIMA(2,1,1)	0.0063	0.73 p.p.	4.97 p.p.	Review residual autocorrelation

The residual checks give a mixed but realistic picture. Earned premium, lapse rate and loss ratio do not show material residual autocorrelation under the Ljung-Box check. Claims count and operating margin are flagged for review. This does not invalidate the project. In a monitoring workflow, the purpose of diagnostics is to identify model risk rather than to force every model to look perfect.

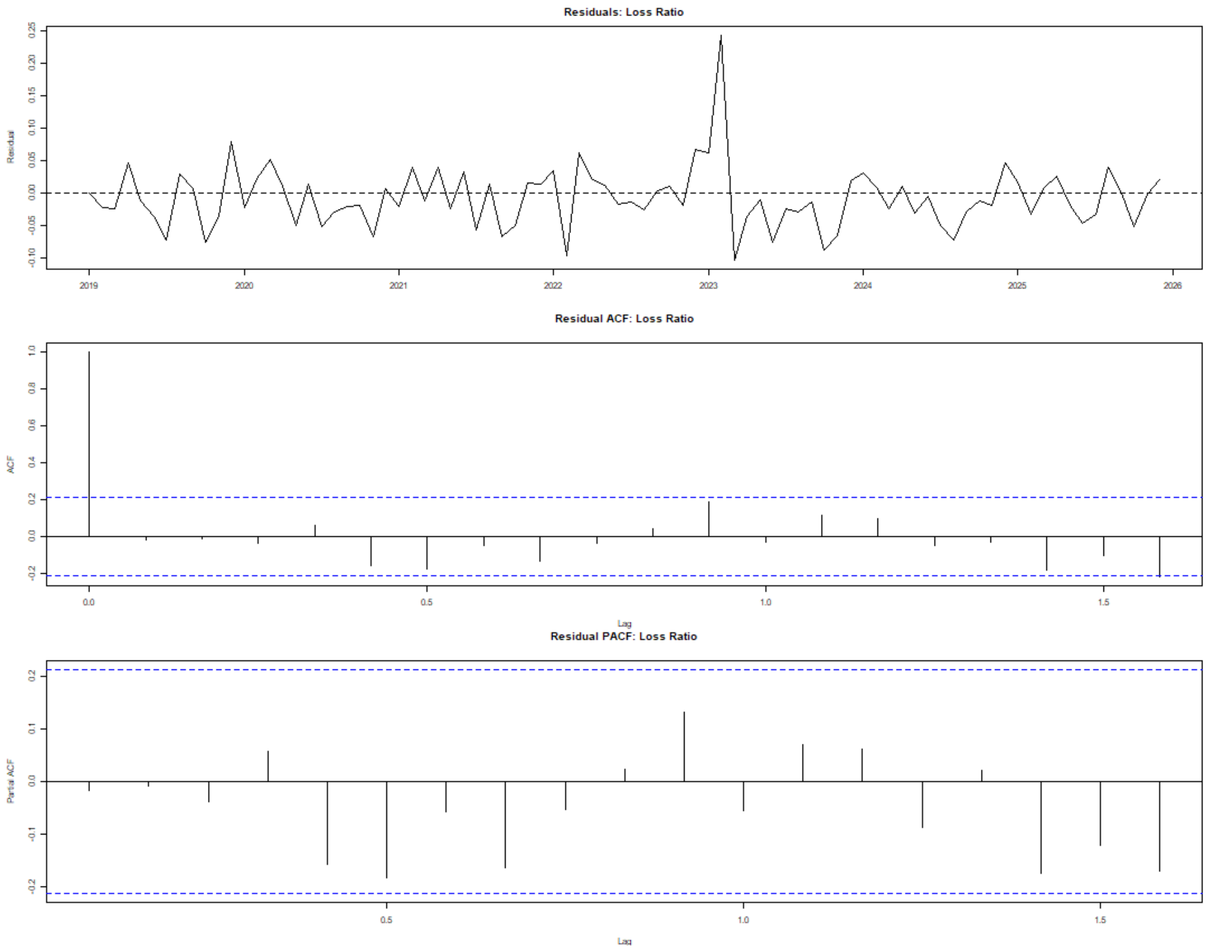

Figure 4: Representative loss-ratio residual diagnostics. The same residual summary process is applied across all five forecasted KPIs, while loss ratio is shown visually because it is a key insurance performance measure.

The loss-ratio diagnostics are used as the representative visual check. This avoids overloading the report with separate residual, ACF and PACF plots for every KPI, while still demonstrating that model health checks were implemented.

7 KPI monitoring and anomaly alerts

The monitoring layer converts the modelling results into a dashboard-style process. It uses two main alert types:

- **Rolling threshold breach:** the current KPI value is compared with a prior 12-month rolling mean and a plus/minus two-standard-deviation band.
- **Forecast interval breach:** the actual value is compared with the one-step-ahead 95% prediction interval.

The rolling threshold uses prior observations only. This avoids look-ahead bias and makes the alert rule closer to a process that could have been applied at the time.

7.1 Current KPI snapshot

The current KPI snapshot reports the final observation date and recent movement across the five KPIs. Volume and revenue measures are reported using relative percentage change. Rate and margin KPIs are reported using percentage-point change because relative percentage changes can be misleading for bounded ratio series.

Table 6: Current KPI snapshot at final observation date

KPI	Date	Current value	MoM change	YoY change	Change basis
Claims Count	2025-12-01	479	-0.2%	-5.5%	Relative percentage change
Earned Premium / Revenue	2025-12-01	1,520,433	+0.3%	+6.3%	Relative percentage change
Lapse Rate	2025-12-01	4.98%	+0.52 p.p.	+0.57 p.p.	Percentage-point change
Loss Ratio	2025-12-01	43.11%	+2.12 p.p.	-2.13 p.p.	Percentage-point change
Operating Margin Proxy	2025-12-01	33.00%	-3.17 p.p.	+0.38 p.p.	Percentage-point change

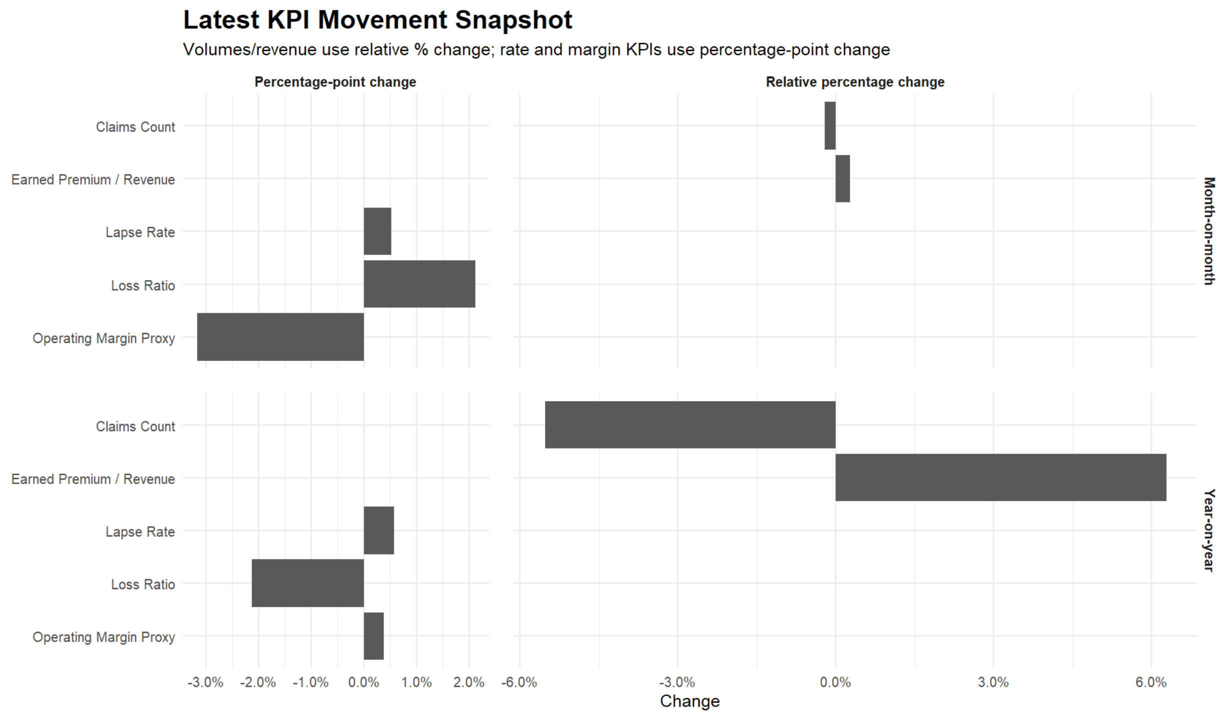

Figure 5: Latest KPI movement snapshot. Volumes and revenue use relative percentage change; rates and margins use percentage-point change.

The snapshot shows why a single movement definition is not appropriate for all KPI types. A 2 percentage-point change in loss ratio or operating margin can be commercially meaningful even if the relative percentage movement is less intuitive.

7.2 Alert summary

The alert summary counts rolling threshold breaches and forecast interval breaches by KPI. This makes the monitoring output easier to read than a long raw alert table.

Table 7: Alert count summary by KPI and alert type

KPI	Alert type	Alert count
Claims Count	Forecast interval breach	2
Claims Count	Rolling threshold breach	4
Earned Premium / Revenue	Forecast interval breach	2
Earned Premium / Revenue	Rolling threshold breach	8
Lapse Rate	Forecast interval breach	1
Lapse Rate	Rolling threshold breach	9
Loss Ratio	Forecast interval breach	2
Loss Ratio	Rolling threshold breach	4
Operating Margin Proxy	Forecast interval breach	2
Operating Margin Proxy	Rolling threshold breach	3

Earned premium and lapse rate have the highest number of rolling threshold alerts. This does not automatically mean the models are poor. It indicates that these KPIs experienced movements outside their recent rolling behaviour and would merit review under the monitoring rules.

Table 8: Ten most recent alerts

Date	KPI	Alert type	Detail	Actual	Reference	Lower threshold	Upper threshold
2025-12-01	Earned Premium / Revenue	Rolling threshold breach	Above threshold	1,520,433	1,425,755	1,337,454	1,514,057
2025-12-01	Lapse Rate	Rolling threshold breach	Above threshold	4.98%	3.94%	3.09%	4.80%
2025-11-01	Earned Premium / Revenue	Rolling threshold breach	Above threshold	1,516,358	1,417,931	1,350,485	1,485,376
2025-05-01	Earned Premium / Revenue	Forecast interval breach	Outside interval	1,390,540	1,339,196	1,291,293	1,387,099
2024-10-01	Earned Premium / Revenue	Forecast interval breach	Outside interval	1,319,876	1,404,779	1,356,876	1,452,682
2024-06-01	Lapse Rate	Rolling threshold breach	Below threshold	3.29%	4.17%	3.35%	4.99%
2024-03-01	Lapse Rate	Forecast interval breach	Outside interval	5.23%	4.17%	3.50%	4.83%
2024-03-01	Lapse Rate	Rolling threshold breach	Above threshold	5.23%	4.22%	3.67%	4.77%
2023-12-01	Earned Premium / Revenue	Rolling threshold breach	Above threshold	1,425,169	1,307,865	1,202,650	1,413,081
2023-11-01	Earned Premium / Revenue	Rolling threshold breach	Above threshold	1,404,575	1,301,517	1,214,071	1,388,962

The detailed alert table is deliberately limited to the ten most recent alerts. The full alert output can be retained as a CSV, but showing every alert in the main report would make the report look like a raw system dump rather than a dashboard.

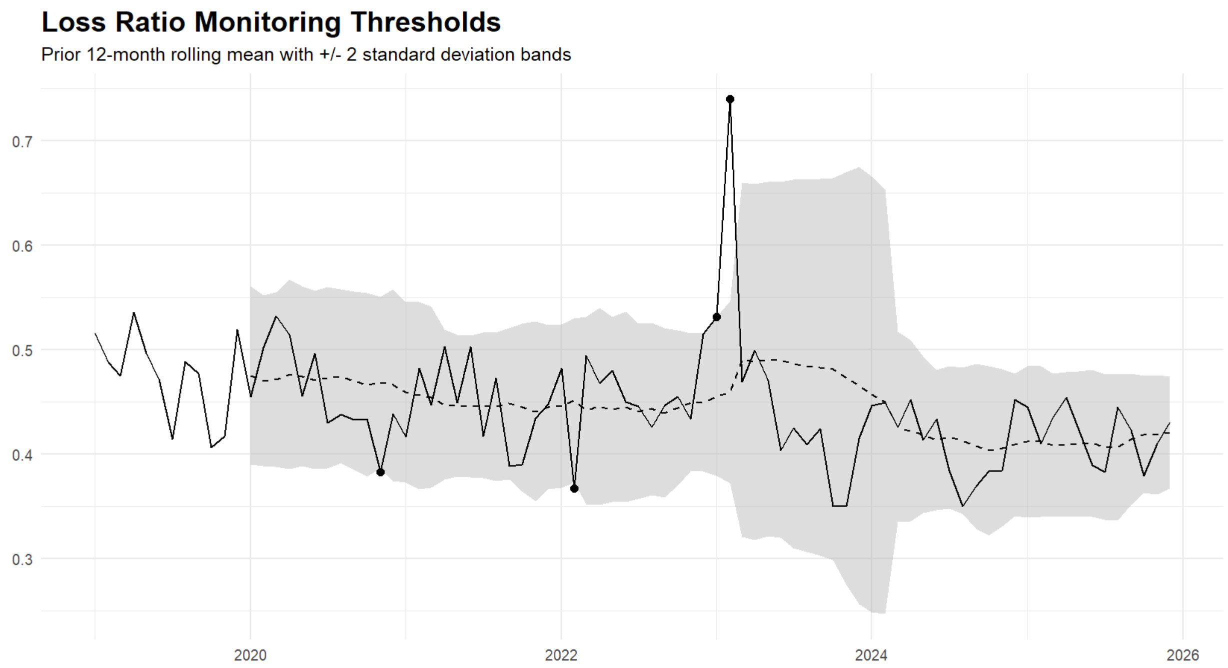

Figure 6: Loss-ratio monitoring against prior 12-month rolling threshold bands. The shaded band represents the rolling mean plus or minus two standard deviations.

The loss-ratio monitoring chart shows how the threshold rule behaves during stable and stressed periods. The band widens after the 2023 shock, which is expected because the prior 12-month standard deviation increases. This is useful from a monitoring perspective because it avoids treating every high value as equally unusual regardless of the recent volatility regime.

8 Interpretation

The project demonstrates a complete KPI forecasting and monitoring workflow rather than a single isolated model. The ARIMA component provides time-series forecasts and prediction intervals. The regression component links KPI movements to interpretable drivers. The rolling evaluation checks whether forecasts are stable out of sample. The residual diagnostics identify model health issues. The alert framework converts forecasts and rolling thresholds into review triggers.

From an insurance analytics perspective, the most relevant KPIs are claims count, loss ratio and lapse rate. Claims count supports claims volume monitoring, loss ratio supports underwriting and pricing performance analysis, and lapse rate supports retention and policyholder behaviour monitoring.

From a finance and business analytics perspective, earned premium and operating margin proxy are more directly relevant. They allow the same project to be discussed as a financial performance forecasting and business analytics workflow. This is why the project can be framed either as *KPI Forecasting and Monitoring Diagnostics* for insurance roles or as *Financial Performance Forecasting and Business Analytics* for broader finance roles.

9 Limitations and further work

The project has several limitations.

- **Synthetic data:** the dataset is designed to be realistic, but it is not a live insurance portfolio. The results should therefore be interpreted as a modelling demonstration rather than a business conclusion.
- **Baseline forecasting models:** ARIMA provides a useful benchmark, but a production system could also consider ARIMAX, dynamic regression, state-space models or hierarchical forecasting.
- **Residual autocorrelation:** claims count and operating margin are flagged for residual autocorrelation. In a production setting, these models would need further refinement.
- **MAPE limitations:** MAPE can be difficult to interpret for rate and margin series. Percentage-point MAE and RMSE are often more useful for these KPIs.
- **Threshold sensitivity:** the alert system uses a prior 12-month mean and plus/minus two standard deviation bands. Other businesses may prefer percentile-based, risk-based or governance-driven thresholds.

Further work could include adding exogenous ARIMAX forecasts, automated recalibration rules, model version tracking, severity-based alert ranking and a more formal model governance section. For insurance use cases, the project could also be extended to include claims severity, combined ratio, reserve development or pricing adequacy indicators.

10 Conclusion

This project builds a compact but complete KPI forecasting and monitoring workflow in R. It uses synthetic insurance and operational KPIs to demonstrate ARIMA forecasting, AIC-based regression modelling, rolling forecast evaluation, residual diagnostics and anomaly alerts.

The results show that the forecasting models produce plausible prediction intervals and useful rolling error metrics. The monitoring layer adds practical value by converting model outputs into KPI movement summaries and alert triggers. The workflow is therefore suitable as a portfolio project for insurance analytics, actuarial analytics, finance analytics and broader business performance monitoring roles.

The project also supports two CV framings. For insurance and actuarial roles, the emphasis is on claims count, loss ratio, lapse monitoring and residual model health checks. For finance and business analytics roles, the emphasis is on earned premium, operating margin, KPI movement reporting and dashboard-style anomaly monitoring.

References

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